

really uses more than 300 W. By capping the amount of power each server can draw to 300 W, we can pack an additional 3333 servers in our datacenter. Of course, power capping and increasing workloads make it challenging to ensure low latency for those tasks that need it, while providing sufficient throughput for batch jobs (Li et al., 2020).

5.10 SUMMARY

Input/output is an often neglected, but important, topic. A substantial fraction of any operating system is concerned with I/O. I/O can be accomplished in one of three ways. First, there is programmed I/O, in which the main CPU inputs or outputs each byte or word and sits in a tight loop waiting until it can get or send the next one. Second, there is interrupt-driven I/O, in which the CPU starts an I/O transfer for a character or word and goes off to do something else until an interrupt arrives signaling completion of the I/O. Third, there is DMA, in which a separate chip manages the complete transfer of a block of data, given an interrupt only when the entire block has been transferred.

I/O can be structured in four levels: the interrupt-service procedures, the device drivers, the device-independent I/O software, and the I/O libraries and spoolers that run in user space. The device drivers handle the details of running the devices and providing uniform interfaces to the rest of the operating system. The device-independent I/O software does things like buffering and error reporting.

Secondary storage comes in a variety of types, including magnetic disks, RAIDs, and flash drives. On rotating disks, disk arm scheduling algorithms can often be used to improve performance, but the presence of virtual geometries complicates matters. By pairing two disks or SSDs, a stable storage medium with certain useful properties can be constructed.

Clocks are used for keeping track of the real time, limiting how long processes can run, handling watchdog timers, and doing accounting.

Character-oriented terminals have a variety of issues concerning special characters that can be input and special escape sequences that can be output. Input can be in raw mode or cooked mode, depending on how much control the program wants over the input. Escape sequences on output control cursor movement and allow for inserting and deleting text on the screen.

Most UNIX systems use the X Window System as the basis of the user interface. It consists of programs that are bound to special libraries that issue drawing commands and an X server that writes on the display.

Many personal computers use GUIs for their output. These are based on the WIMP paradigm: windows, icons, menus, and a pointing device. GUI-based programs are generally event driven, with keyboard, mouse, and other events being sent to the program for processing as soon as they happen. In UNIX systems, the GUIs almost always run on top of X.

Thin clients have some advantages over standard PCs, notably simplicity and less maintenance for users.

Finally, power management is a major issue for phones, tablets, and notebooks because battery lifetimes are limited and for desktop and server machines because of an organization's energy bills. Various techniques can be employed by the operating system to reduce power consumption. Programs can also help out by sacrificing some quality for longer battery lifetimes.

PROBLEMS

1. Advances in chip technology have made it possible to put an entire controller, including all the bus access logic, on an inexpensive chip. How does that affect the model of Fig. 1-6?
2. Given the speeds listed in Fig. 5-1, is it possible to record video using a digital video recorder and transmit them over an 802.11n network at full speed? Defend your answer.
3. Figure 5-3(b) shows one way of having memory-mapped I/O even in the presence of separate buses for memory and I/O devices, namely, to first try the memory bus and if that fails try the I/O bus. A clever computer science student has thought of an improvement on this idea: try both in parallel, to speed up the process of accessing I/O devices. What do you think of this idea?
4. A DMA controller has four channels. The controller is capable of requesting a 32-bit word every 100 nsec. A response takes equally long. How fast does the bus have to be to avoid being a bottleneck?
5. Suppose that a system uses DMA for data transfer from disk controller to main memory. Further assume that it takes t_1 nsec on average to acquire the bus and t_2 nsec to transfer one word over the bus ($t_1 \gg t_2$). After the CPU has programmed the DMA controller, how long will it take to transfer 1000 words from the disk controller to main memory, if (a) word-at-a-time mode is used, (b) burst mode is used? Assume that commanding the disk controller requires acquiring the bus to send one word and acknowledging a transfer also requires acquiring the bus to send one word.
6. One mode that some DMA controllers use is to have the device controller send the word to the DMA controller, which then issues a second bus request to write to memory. How can this mode be used to perform memory to memory copy? Discuss any advantage or disadvantage of using this method instead of using the CPU to perform memory to memory copy.
7. Explain the difference among interrupts, exceptions/faults, and traps with concrete examples.
8. Suppose that a computer can read or write a memory word in 10 nsec. Also suppose that when an interrupt occurs, all 32 CPU registers, plus the program counter and PSW are pushed onto the stack. What is the maximum number of interrupts per second this machine can process?

9. In Fig. 5-9(b), the interrupt is not acknowledged until after the next character has been output to the printer. Could it have equally well been acknowledged right at the start of the interrupt service procedure? If so, give one reason for doing it at the end, as in the text. If not, why not?
10. A computer has a three-stage pipeline as shown in Fig. 1-7(a). On each clock cycle, one new instruction is fetched from memory at the address pointed to by the PC and put into the pipeline and the PC advanced. Each instruction occupies exactly one memory word. The instructions already in the pipeline are each advanced one stage. When an interrupt occurs, the current PC is pushed onto the stack, and the PC is set to the address of the interrupt handler. Then the pipeline is shifted right one stage and the first instruction of the interrupt handler is fetched into the pipeline. Does this machine have precise interrupts? Defend your answer.
11. For some applications, a typical printed page of text contains 45 lines of 80 characters each. Imagine that a certain printer can print 6 pages per minute and that the time to write a character to the printer's output register is so short it can be ignored. Does it make sense to run this printer using interrupt-driven I/O if each character printed requires an interrupt that takes 50 μ sec all-in to service?
12. Explain how an OS can facilitate installation of a new device without any need for recompiling the OS.
13. In which of the four I/O software layers is each of the following done?
 - (a) Computing the track, sector, and head for a disk read.
 - (b) Writing commands to the device registers.
 - (c) Checking to see if the user is permitted to use the device.
 - (d) Converting binary integers to ASCII for printing.
14. A local area network is used as follows. The user issues a system call to write data packets to the network. The operating system then copies the data to a kernel buffer. Then it copies the data to the network controller board. When all the bytes are safely inside the controller, they are sent over the network at a rate of 10 megabits/sec. The receiving network controller stores each bit a microsecond after it is sent. When the last bit arrives, the destination CPU is interrupted, and the kernel copies the newly arrived packet to a kernel buffer to inspect it. Once it has figured out which user the packet is for, the kernel copies the data to the user space. If we assume that each interrupt and its associated processing takes 1 msec, that packets are 1024 bytes (ignore the headers), and that copying a byte takes 1 μ sec, what is the maximum rate at which one process can pump data to another? Assume that the sender is blocked until the work is finished at the receiving side and an acknowledgement comes back. For simplicity, assume that the time to get the acknowledgement back is so small it can be ignored.
15. Why are output files for the printer normally spooled on disk before being printed?
16. The I/O time of hard disks mainly consists of three parts.
 - (a) Seek time
 - (b) Boot time
 - (c) Rotational delay (d) Actual data transfer time

Which one is the dominant factor in a typical hard disk? What about an SSD?

17. How much cylinder skew is needed for a 7200-RPM disk with a track-to-track seek time of 1 msec? The disk has 1000 sectors of 512 bytes each on each track.
18. A disk rotates at 7200 RPM. It has 200 sectors of 512 bytes around the outer cylinder. How long does it take to read a sector?
19. Calculate the maximum data rate in bytes/sec for the disk described in the previous problem.
20. RAID level 3 is able to correct single-bit errors using only one parity drive. What is the point of RAID level 2? After all, it also can only correct one error and takes more drives to do so.
21. A RAID can fail if two or more of its drives crash within a short time interval. Suppose that the probability of one drive crashing in a given hour is p . What is the probability of a k -drive RAID failing in a given hour?
22. Compare RAID level 0 through 5 with respect to read performance, write performance, space overhead, and reliability.
23. How many pebibytes are there in a zebibyte?
24. Why are optical storage devices inherently capable of higher data density than magnetic storage devices? *Note:* This problem requires some knowledge of high-school physics and how magnetic fields are generated.
25. What are the advantages and disadvantages of optical disks versus magnetic disks?
26. If a disk controller writes the bytes it receives from the disk to memory as fast as it receives them, with no internal buffering, is interleaving conceivably useful? Discuss.
27. If a disk has double interleaving, does it also need cylinder skew in order to avoid missing data when making a track-to-track seek? Discuss your answer.
28. Consider a magnetic disk consisting of 16 heads and 400 cylinders. This disk has four 100-cylinder zones with the cylinders in different zones containing 160, 200, 240, and 280 sectors, respectively. Assume that each sector contains 512 bytes, average seek time between adjacent cylinders is 1 msec, and the disk rotates at 7200 RPM. Calculate the (a) disk capacity, (b) optimal track skew, and (c) maximum data transfer rate.
29. A disk manufacturer has two 3.5-inch disks that each have 15,000 cylinders. The newer one has double the linear recording density of the older one. Which disk properties are better on the newer drive and which are the same?
30. Suppose some clever computer science student decides to redesign the MBR and partition table of a hard disk to provide more than four partitions. What are some consequences of this change?
31. Disk requests come in to the disk driver for cylinders 10, 22, 20, 2, 40, 6, and 38, in that order. A seek takes 6 msec per cylinder. How much seek time is needed for
 - (a) First-come, first served.
 - (b) Closest cylinder next.
 - (c) Elevator algorithm (initially moving upward).

In all cases, the arm is initially at cylinder 20.

32. A slight modification of the elevator algorithm for scheduling disk requests is to always scan in the same direction. In what respect is this modified algorithm better than the elevator algorithm?
33. To improve the I/O performance of hard disks, many scheduling algorithms have been proposed for handling I/O requests, such as FCFS (First-Come, First-Served), SSF (Shortest Seek First), and the elevator algorithm. Which one(s) make the most sense for SSDs? Explain your answer.
34. Compared to hard disks, SSDs are different in some ways. Which of the following statements about SSDs are true?
- (a) SSDs can handle more I/O requests in parallel.
 - (b) SSDs do not incur rotational delay.
 - (c) SSDs are more resilient to vibration because they contain no moving parts.
 - (d) SSDs do not incur seek time.
 - (e) SSDs are cheaper per megabyte.
35. A personal computer salesman visiting a university in South-West Amsterdam remarked during his sales pitch that his company had devoted substantial effort to making their version of UNIX very fast. As an example, he noted that their disk driver used the elevator algorithm and also queued multiple requests within a cylinder in sector order. A student, Harry Hacker, was impressed and bought one. He took it home and wrote a program to randomly read 10,000 blocks spread across the disk. To his amazement, the performance that he measured was identical to what would be expected from first-come, first-served. Was the salesman lying?
36. In the discussion of stable storage using nonvolatile RAM, the following point was glossed over. What happens if the stable write completes but a crash occurs before the operating system can write an invalid block number in the nonvolatile RAM? Does this race condition ruin the abstraction of stable storage? Explain your answer.
37. In the discussion on stable storage, it was shown that the disk can be recovered to a consistent state (a write either completes or does not take place at all) if a CPU crash occurs during a write. Does this property hold if the CPU crashes again during a recovery procedure. Explain your answer.
38. In the discussion on stable storage, a key assumption is that a CPU crash that corrupts a sector leads to an incorrect ECC. What problems might arise in the five crash-recovery scenarios shown in Figure 5-27 if this assumption does not hold?
39. The clock interrupt handler on a certain computer requires 2 msec (including process switching overhead) per clock tick. The clock runs at 60 Hz. What fraction of the CPU is devoted to the clock?
40. A computer uses a programmable clock in square-wave mode. If a 1 GHz crystal is used, what should be the value of the holding register to achieve a clock resolution of
- (a) a millisecond (a clock tick once every millisecond)?
 - (b) 100 microseconds?
41. A system simulates multiple clocks by chaining all pending clock requests together as shown in Fig. 5-29. Suppose the current time is 5000 and there are pending clock

- requests for time 5008, 5012, 5015, 5029, and 5037. Show the values of Clock header, Current time, and Next signal at times 5000, 5005, and 5013. Suppose a new (pending) signal arrives at time 5017 for 5033. Show the values of Clock header, Current time and Next signal at time 5023.
42. Many versions of UNIX use an unsigned 32-bit integer to keep track of the time as the number of seconds since the origin of time. When will these systems wrap around (year and month)? Do you expect this to actually happen?
 43. Consider the performance of a 56-kbps modem of yesteryear (which are still common in rural areas without broadband). The driver outputs one character and then blocks. When the character has been printed, an interrupt occurs and a message is sent to the blocked driver, which outputs the next character and then blocks again. If the time to pass a message, output a character, and block is $100\ \mu\text{sec}$, what fraction of the CPU is eaten by the modem handling? Assume that each character has one start bit and one stop bit, for 10 bits in all.
 44. A smartphone screen contains 720×1280 pixels. To scroll a full screen of text, the CPU (or controller) must move all the lines of text upward by copying their bits from one part of the video RAM to another. A character's box is 16 pixels wide by 32 pixels high (including intercharacter and interline spacing). Each pixel is 24 bits. How many characters fit on the screen? How long does it take to scroll the whole screen at a copying rate of 5 nsec per byte assuming there is no hardware assistance? What is the scrolling speed in lines/sec?
 45. After receiving a DEL (SIGINT) character, the display driver discards all output currently queued for that display. Why?
 46. A user issues a command to an editor to delete the word on line 5 occupying character positions 7 through and including 12. Assuming the cursor is not on line 5 when the command is given, what ANSI escape sequence should the editor emit to delete the word?
 47. The designers of a computer system expected that the mouse could be moved at a maximum rate of 20 cm/sec. If a mickey is 0.1 mm and each mouse message is 3 bytes, what is the maximum data rate of the mouse assuming that each mickey is reported separately?
 48. The primary additive colors are red, green, and blue, which means that any color can be constructed from a linear superposition of these colors. Is it possible that someone could have a color photograph that cannot be represented using full 24-bit color?
 49. One way to place a character on a bitmapped screen is to use *BitBlt* from a font table. Assume that a particular font uses characters that are 16×24 pixels in true RGB color.
 - (a) How much font table space does each character take?
 - (b) If copying a byte takes 100 nsec, including overhead, what is the output rate to the screen in characters/sec?
 50. Assuming that it takes 5 nsec to copy a byte, how much time does it take to completely rewrite the screen of an 80 character $\times$ 25 line text mode memory-mapped screen? What about a 1536×2048 pixel graphics screen with 24-bit color?

51. In Fig. 5-36 there is a class to *RegisterClass*. In the corresponding X Window code, in Fig. 5-34, there is no such call or anything like it. Why not?
52. In the text we gave an example of how to draw a rectangle on the screen using the Windows GDI:

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Rectangle(hdc, xleft, ytop, xright, ybottom);
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Is there any real need for the first parameter (*hdc*), and if so, what? After all, the coordinates of the rectangle are explicitly specified as parameters.

53. A thin-client terminal is used to display a Web page containing an animated cartoon of size 400 pixels $\times$ 160 pixels running at 20 frames/sec. What fraction of a 1000-Mbps gigabit Ethernet is consumed by displaying the cartoon?
54. It has been observed that a thin-client system works well with a 1-Mbps network in a test. Are any problems likely in a multiuser situation? (*Hint*: Consider a large number of users watching a scheduled TV show and the same number of users browsing the World Wide Web.)
55. Describe two advantages and two disadvantages of thin client computing.
56. If a CPU's maximum voltage, V , is cut to V/n , its power consumption drops to $1/n^2$ of its original value and its clock speed drops to $1/n$ of its original value. Suppose that a user is typing at 1 char/sec, but the CPU time required to process each character is 100 msec. What is the optimal value of n and what is the corresponding energy saving in percent compared to not cutting the voltage? Assume that an idle CPU consumes no energy at all.
57. A notebook computer is set up to take maximum advantage of power saving features including shutting down the display and the hard disk after periods of inactivity. A user sometimes runs UNIX programs in text mode, and at other times uses the X Window System. She is surprised to find that battery life is significantly better when she uses text-only programs. Why?
58. Write a program that simulates stable storage. Use two large fixed-length files on your disk to simulate the two disks.
59. Write a program to implement the three disk-arm scheduling algorithms. Write a driver program that generates a sequence of cylinder numbers (0–999) at random, runs the three algorithms for this sequence and prints out the total distance (number of cylinders) the arm needs to traverse in the three algorithms.
60. Write a program to implement multiple timers using a single clock. Input for this program consists of a sequence of four types of commands (S <int>, T, E <int>, P): S <int> sets the current time to <int>; T is a clock tick; and E <int> schedules a signal to occur at time <int>; P prints out the values of Current time, Next signal, and Clock header. Your program should also print out a statement whenever it is time to raise a signal.